

3.6.1 Periodic motion (A-level only)

3.6.1.1 Circular motion (A-level only)

Content	Opportunities for skills development
Motion in a circular path at constant speed implies there is an acceleration and requires a centripetal force. Magnitude of angular speed $\omega = \frac{v}{r} = 2\pi f$ Radian measure of angle. Direction of angular velocity will not be considered. Centripetal acceleration $a = \frac{v^2}{r} = \omega^2 r$ The derivation of the centripetal acceleration formula will not be examined. Centripetal force $F = \frac{mv^2}{r} = m\omega^2 r$	MS 0.4 Estimate the acceleration and centripetal force in situations that involve rotation.

3.6.1.2 Simple harmonic motion (SHM) (A-level only)

Content	Opportunities for skills development
Analysis of characteristics of simple harmonic motion (SHM). Condition for SHM: $a \propto -x$ Defining equation: $a = -\omega^2 x$ $x = A \cos \omega t$ and $v = \pm \omega \sqrt{(A^2 - x^2)}$ Graphical representations linking the variations of x, v and a with time. Appreciation that the $v - t$ graph is derived from the gradient of the $x - t$ graph and that the $a - t$ graph is derived from the gradient of the $v - t$ graph. Maximum speed = ωA Maximum acceleration = $\omega^2 A$	AT i, k Data loggers can be used to produce $s - t$, $v - t$ and $a - t$ graphs for SHM. MS 3.6, 3.8, 3.9, 3.12 Sketch relationships between x, v, a and t for simple harmonic oscillators.

3.6.1.3 Simple harmonic systems (A-level only)

Content	Opportunities for skills development
Study of mass-spring system: $T = 2\pi\sqrt{\frac{m}{k}}$ Study of simple pendulum: $T = 2\pi\sqrt{\frac{l}{g}}$ Questions may involve other harmonic oscillators (eg liquid in U-tube) but full information will be provided in questions where necessary. Variation of E_k, E_p, and total energy with both displacement and time. Effects of damping on oscillations.	MS 4.6 / AT b, c Students should recognise the use of the small-angle approximation in the derivation of the time period for examples of approximate SHM.
Required practical 7: Investigation into simple harmonic motion using a mass–spring system and a simple pendulum.	

3.6.1.4 Forced vibrations and resonance (A-level only)

Content	Opportunities for skills development
Qualitative treatment of free and forced vibrations. Resonance and the effects of damping on the sharpness of resonance. Examples of these effects in mechanical systems and situations involving stationary waves.	AT g, i, k Investigation of the factors that determine the resonant frequency of a driven system.